

DEVENDER KOMATI REDDY

Lead Data Scientist /Agentic AI Engineer

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PROFESSIONAL SUMMARY

- An Innovative **Lead Data Scientist/Machine Learning Engineer with 10+ years** of experience in developing, deploying, and leading machine learning and big data projects and **Agentic AI Engineer with 2+ years** of experience in LangGraph, LangChain, Retrieval-Augmented Generation (RAG), vector databases (Pinecone, FAISS, Weaviate, Milvus), SQL, JSON schema, function calling, and tool invocation.
- Specialized in NLP, Computer Vision, and PySpark, with hands-on experience in building POCs using LangChain, LangGraph and Llama. Strong expertise in LLMs, Agentic AI, MLOps and cloud computing, backed by a solid track record of successful project deliveries.
- Certified Data Scientist, AI Engineer and Solution Architect Expert from Microsoft Azure.
- Expert in using the ML platforms PyTorch and TensorFlow to solve problems in Computer Vision, Natural Language Processing. Efficient in data preprocessing, including Data cleaning, Correlation analysis, Imputation, Visualization, Feature Scaling, and Dimensionality Reduction techniques using Machine learning platforms namely Python Data Science Packages (Scikit-Learn, Pandas, NumPy).
- Skilled in fine-tuning and deploying Large Language Models (LLMs) to customize outputs for specific use cases, including text summarization, information retrieval, and sentiment analysis, delivering high-quality, contextually accurate responses.
- Proficient in Retrieval-Augmented Generation (RAG) techniques, combining LLMs with vector databases for enhanced response accuracy by retrieving relevant contextual data in real-time.
- Experienced in prompt engineering and optimization, crafting effective prompts to refine LLM outputs for domain-specific applications, while maximizing performance and maintaining efficiency.
- Demonstrated expertise in integrating LLMs with cloud services for scalable deployment, enabling rapid development and testing cycles while maintaining model performance at scale.
- Strong experience in Statistical Analysis, Survey Analytics, Regression, Classification, Clustering and Time Series Analysis & Forecasting.
- Experience in applying Classification algorithms like Logistic Regression, K-Nearest Neighbors, SVM, Decision Trees and Naïve Bayes algorithm, Ensemble Classifiers: Bagging, Random Forest, Gradient Boosting (XGBoost)
- Experience in applying Regression algorithms like Linear Regression, Ridge, Lasso and Elastic Net Regression, SVR, Ensemble Regressors: Gradient Boosting, Random Forest Experience in applying Segmentation/Clustering algorithms like K-Means clustering, Hierarchical clustering Experience in applying Dimensionality Reduction techniques like Principal Component Analysis (PCA)
- Strong experience in Data Processing, Data transformation, Data quality check using Python and PySpark on Databricks platform

- Experienced in creating data visualizations using plotly, pyplot and seaborn also proficient in Power BI, Fabric, Tableau, SAP Business Intelligence data
- Experienced in Model Development and Registry, Model Deployment and Monitoring and Governance.
- Hands on experience on AWS services like Sage Maker, S3, Athena, Lambda, API Gateway, Glue, EMR, EC2, DynamoDB, Redshift, AWS CLI, ECR, Step Functions and CloudWatch.
- Hands on experience on GCP services like Vertex AI, Big Query, GCP Buckets, Kubeflow, Airflow, Cloud Functions and VM Instances
- Extracted data from various data sources like Oracle, SQL Server, DB2, Teradata, Snowflake, Vertica and NoSQL databases (MongoDB)
- Hands on MLOps Tools like AWS Sage maker, Vertex AI, GitHub, Docker, Kubernetes, Kubeflow, Airflow and Terraform.

PROFILES

- <https://github.com/devenderreddy26>
- <https://www.linkedin.com/in/devender-komati-reddy/>

SKILLS

- **Programming Languages:** Python (PySpark, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, OOP), Scala, SQL, Java, FastAPI, REST APIs, Git
- **Cloud Platforms:** Azure, GCP, AWS
- **Deep Learning:** ANN, RBM, DBN, CNN, RNN; Experience with Keras API and AWS Deep Learning
- **AI Development:** LangChain, LangGraph, Multi-agent workflows, Vector DBs (FAISS, Pinecone, Milvus, Weaviate), LLM fine-tuning, Retrieval-Augmented Generation (RAG), GraphRAG, prompt engineering, model optimization, Generative AI techniques, JSON schema, function calling, input/output guardrails, tool invocation, Azure AI Foundry, Llama, OpenAI
- **Databases:** SQL (Postgres, MySQL, Oracle, Redshift), NoSQL (MongoDB, Redis), ADLS
- **MLOps:** AWS SageMaker, Vertex AI, CI/CD with GitHub Actions, Jenkins, GitLab CI, AWS Code Pipeline; containerization (Docker, Kubernetes), model monitoring and logging (CloudWatch, Grafana, Prometheus)
- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, PySpark, Keras; Algorithms: Linear & Logistic Regression, Naive Bayes, Decision Trees, Random Forest, SVM, K-Means, KNN, Gradient & Ada Boosting, PCA, LDA, Sentiment Analysis, NLP

CERTIFICATIONS

AI Agent Developer from Vanderbilt University 06/2025

Microsoft Certified: Azure Solutions Architect Expert 06/2025

Microsoft Certified: Azure AI Engineer Associate 05/2025

Large Language Model Operations (LLMOps) from Duke University 03/2024

Generative AI with Large Language Models from DeepLearning.AI 12/2023

Deep Learning Specialization from DeepLearning.AI 03/2023

LLM101x: Large Language Models: Application through Production from edX 03/2024

Microsoft Certified: Azure Data Scientist Associate 11/2022

WORK HISTORY

AGENTIC AI ENGINEER / LEAD DATA SCIENTIST | Wells Fargo - Chandler, AZ 09/2023 - Current

- Designed and deployed agentic AI pipelines using LangGraph, LangChain and multi-agent orchestration to automate loan processing, compliance workflows, and customer service—reducing manual intervention by 35%.
- Implemented Retrieval-Augmented Generation (RAG and GraphRAG) systems combining vector DBs (Pinecone), SQL metadata stores, and hybrid search to achieve 85%+ retrieval accuracy on regulatory documents.
- Developed structured output pipelines (JSON schema, tool/function calling) with I/O validation guardrails, ensuring compliance with financial regulations and data security policies.
- Designed evaluation frameworks with golden queries and regression testing to benchmark RAG accuracy, latency, and hallucination rates across model updates
- Delivered scalable AI infrastructure leveraging CI/CD pipelines, Docker, Kubernetes, and monitoring tools, ensuring high availability and compliance in regulated environments.
- Partnered with fraud, compliance, and onboarding teams to pilot next-gen AI solutions in fraud detection, AML monitoring, and digital onboarding, driving enterprise-wide adoption of generative AI.
- Built a production RAG service using Prompt flow, Azure AI Search (vector), and managed endpoints in Azure AI Foundry; added evals/telemetry to cut hallucinations
- Designed Lakehouse centric ML pipelines in OneLake using Notebooks + Dataflows Gen2; trained/tracked models with MLflow and automated runs with Fabric pipelines.
- Used various Parameter-efficient fine-tuning (PEFT) techniques like LoRA, QLoRA and GRPO to fine-tune the foundation models.
- Evaluated the LLMs using traditional metrics like BLEU, ROGUE and BERT score and state-of-the-art approaches like LLM-based evaluation.
- Demonstrated proficiency in working with indexing and similarity search engines (Vector DBs) such as Milvus, Pinecone, and Chroma DB to enhance data retrieval and pattern recognition capabilities.

LEAD DATA SCIENTIST | Wells Fargo - Chandler, AZ

01/2016 – 09/2023

- Led a team of data scientists and engineers in developing credit risk assessment model that reduced default rates by 24%, using H2O XGBoost, and contributed to the development of a fraud detection system. Added 60+ new features improving AUC score by 23%
- Established model training pipelines on Azure Databricks for data preprocessing, reducing 50% processing time and 70% compute and memory usage

- Participated in all phases of data mining, data collection, data cleaning, developing models, validation and visualization and performed gap analysis using Python and PySpark.
- Worked on Azure Databricks and used Data Frame API, built-in functions, user-defined functions in PySpark and Pandas for data preprocessing
- Performed data analysis, data wrangling and statistical analysis with complex large datasets to proactively identify opportunities for cost and utilization improvements.
- Led projects to establish data pipelines to automatically extract, wrangle and manipulate unstructured data and reduced 40% lead time.
- Trained ATO models to prevent account takeover activity at the time of user authentication
- Assisted in building a customer segmentation model using k-means clustering that resulted in 24% increase in targeted marketing campaign conversion rate
- Used Pandas, NumPy, seaborn, SciPy, Matplotlib, Scikit-learn, NLTK and Python in Jupyter notebooks for various machine learning algorithms and utilized machine learning algorithms such as Logistic Regression, linear regression, Random Forests, K-means, & KNN for data analysis.
- Processed data from MongoDB (NoSQL database), Vertica, Snowflake, Teradata and Azure data lake (ADLS) environments using Python.
- Utilized various techniques like Histogram, Bar-plot, Pie-chart, Scatter Plot, Box plots.
- Conducted data exploration to look for trends, patterns, grouping and deviations in data to understand the data diagnostics. Heavily used SQL along with PySpark for data analysis.
- Performed EDA on more than 2000 features to understand the intrinsic effect/combined effects.
- Balanced the class imbalance in dataset using under-sampling with ensemble methods, over-sampling with SMOTE and cost sensitive algorithms with Scikit-learn.
- Reduced time spent on manual tasks by developing automated processes using Python scripts and SQL queries.
- Managed a team of junior data scientists, fostering talent development and consistent project success.
- Evaluated emerging technologies and tools in the field of data science, recommending strategic investments for the organization.
- Used common data science toolkits such as Python(including OOP), NumPy, Keras, Theano, TensorFlow etc.
- Used F1-Score, Precision, Recall, Confusion matrices and ROC curves to validate Machine Learning classifiers.
- Developed RESTful APIs and microservices-using FastAPI to expose AI model endpoints for downstream integration with enterprise business systems.
- Worked on Dockers and Kubernetes services on Azure platform
- Containerized AI applications using Docker and deployed them on Kubernetes clusters for scalable, fault-tolerant model serving by collaborating with DevOps team
- Built end-to-end MLOps pipelines for continuous integration and deployment (CI/CD) using GitHub Actions and Azure DevOps, reducing manual deployment time by 70%.
- Worked on MLFlow to automate the model deployment

SR BUSINESS INTELLIGENCE ARCHITECT | Wells Fargo - Chandler, AZ

05/2015 - 01/2016

- Optimized ETL processes for faster and more accurate data integration.
- Analyzed complex datasets to identify trends, patterns, and opportunities for growth within the organization.
- Conducted thorough performance evaluations of BI tools, recommending upgrades or replacements as necessary.

- Championed data-driven decision-making throughout the organization by sharing success stories and best practices.
- Developed custom dashboards and reports, providing valuable insights for executive leadership.
- Evaluated emerging technologies within the BI space for potential implementation in current workflows or future projects.
- Created project and application architecture deliverables consistent with architecture principles, standards, methodologies and best practices.
- Developed database architectural strategies at modeling, design, and implementation stages to address business or industry requirements.

SR BUSINESS INTELLIGENCE ARCHITECT | Avnet Inc. - Phoenix, AZ

02/2012 - 05/2015

- Implemented data warehouse structures, ensuring efficient storage and retrieval of critical information.
- Spearheaded the adoption of a new BI platform, leading to increased efficiency in data analysis and reporting tasks.

SAP BUSINESS OBJECTS ADMINISTRATOR | Advanced Energy Industries Inc. - Fort Collins, CO

10/2009 - 02/2012

SAP NETWEAVER ARCHITECT | Astellas Pharmaceuticals - Deerfield, IL

06/2007 - 10/2009

SAP PORTAL CONSULTANT | Amgen Inc - Longmont, CO

09/2006 - 05/2007

SAP PORTAL CONSULTANT | Conoco Philips - Bartlesville, OK

04/2006 - 09/2006

SAP PORTALS CONSULTANT | Infosys Technologies Ltd - Bangalore

06/2004 - 01/2006

JAVA DEVELOPER | CMC Limited - Hyderabad

11/2003 - 06/2004

EDUCATION

Osmania University - Hyderabad, Telangana

Bachelor of Technology: Computer Science & Engineering

06/1999 - 05/2003

Achieved 211 rank in Engineering entrance exam(EAMCET), among 350,000 candidates in our state.

Achieved 2nd rank out of 30,000 students in SCI-MAT (Science and Mathematics) Talent Test